Campus Occupancy

MTM4681 Introduction to Mobile Programming · Milestone 1

Live crowd levels for the cafés, dining hall and gym on YTU Davutpaşa Campus

Team	22
Members	Oğuzhan Arslan · Sıla Bayır · Rabia Şahin
GitHub repository	https://github.com/Ozi123arslan/kampus-doluluk
Figma mockup	https://www.figma.com/design/IMpJjh3kscYoXShQRoBfCW
Date	11 November 2026



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1. Problem and Users

People who use the YTÜ campus cannot know how crowded a café, the dining hall or the gym is before they go. They have to walk there and turn back, wait in long lines, or message friend groups asking "is it full?" and wait for a reply. The time lost looking for a place becomes valuable especially when time is limited: for example, students who want to do a final review with friends before an exam lose time they could spend studying while walking from place to place looking for a seat.

Who experiences it: Students on campus; especially those looking for a seat at Hokka Cafe, Kayıntı Cafe, Yıldız Cafe and Büyük Ev between classes, those using the dining hall for lunch and dinner, those going to the gym, and those joining the line in front of Yirmibirco.

How often: Four out of five students said that in the last one to two weeks they went to a place 2 to 5 times and had to turn back or move somewhere else because it was full.

Evidence: Written interviews (7 questions) were conducted on 09-10.11.2026 with students who use the campus.

- **Turning back and searching for a seat:** Most of the turning back happens at Hokka Cafe and Büyük Ev. Four out of five students said they lose between 15 minutes and 1

hour each time they search for a seat; in the end they move to another place, an empty classroom in a faculty building, or a group study room in the library. User 4: "Walking around all the faculties and campus venues looking for a free spot costs at least an hour, and in the end we go back to the dorm."

- **Weather:** Two students said that on rainy days the number of available seats at Büyük Ev drops noticeably and that they turned back on those days.
- **Lines and peak hours:** According to students' estimates, the line at Yirmibirco reaches 20 to 50 people at peak hours and the wait takes 10-20 minutes; all of the students said they have given up after seeing the line. The dining hall is most crowded at lunchtime, roughly between 12:00 and 13:30; students said that after the long line they end up eating late or go to another place instead. The two students who use the gym said the mixed sessions are very crowded in the afternoon and even reach the capacity limit; as a result they either cut time from their classes to go at quieter hours or give up using the gym.
- **Current method:** Four out of five students have no reliable way to find out how crowded a place is before going; they either go and check or ask through WhatsApp groups and friends. User 1: "We keep in touch with friends in our WhatsApp group, but it doesn't really work because we all check at the same time." One student said that asking friends usually works.
- **Demand:** All of the students said they would use such an app; four said they would report the crowd level while at a venue, and one said they would only check it.

2. Why It Matters: Why Existing Solutions Fall Short

None of the existing channels show the current crowd level of campus venues in one place and in real time; as a result, users only find out once they arrive.

Existing solution	What it does	Why it falls short
YTÜ Mobil	Dining hall menu, campus shuttle times, announcements	No real-time crowd or line information
Google Maps "Popular times"	General crowd estimate based on historical data	Often does not distinguish small campus venues; data is either missing or not live
WhatsApp / Telegram groups	Students ask "is the dining hall full?"	Replies may not come or come late, messages are not searchable, each department has its own group
Cafés' social media accounts	Campaigns, opening and closing announcements	One-way; gives no information about the current crowd
Gym website	Opening hours and announcements	Static information; does not show how crowded the gym is right now
Going and checking	Accurate information	Wastes time; if it's full, you have to walk to another place

What makes us different: Campus cafés, the dining hall, the gym and the line at Yirmibirco on one screen; crowd levels are calculated live from users' current reports, and the number and time of reports are visible. The app also shows the last known status when offline.

Screenshots



Figure 1. YTÜ Mobil – Home: news, daily menu and shortcuts

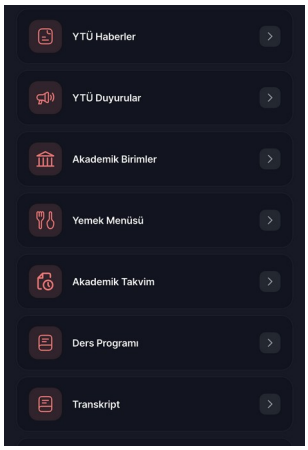


Figure 2. YTÜ Mobil – Menu: news, announcements, meal menu, calendar, timetable



Figure 3. YTÜ Mobil – Daily menu: dining hall menu, no crowd information

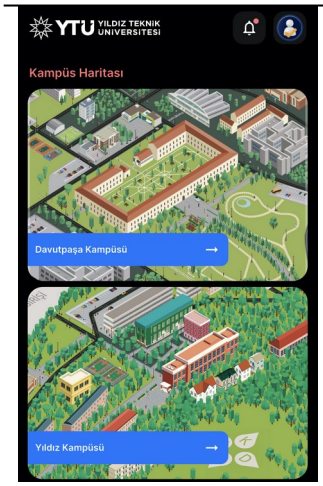


Figure 4. YTÜ Mobil – Campus map

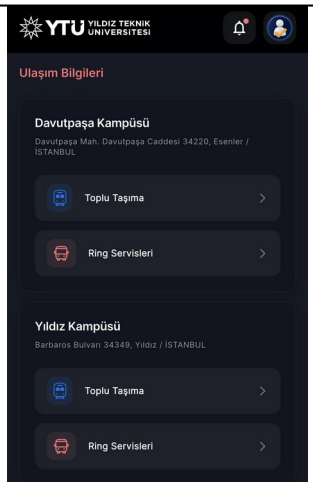


Figure 5. YTÜ Mobil – Transportation and campus shuttles



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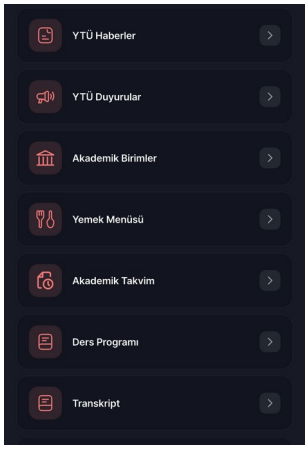


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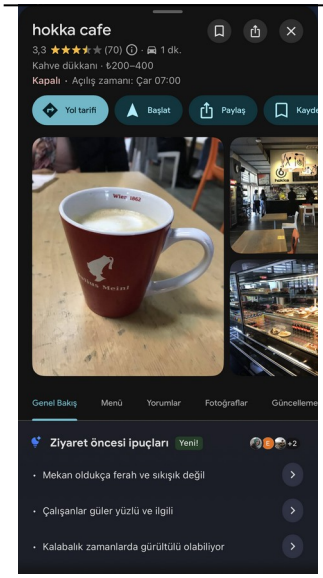


Figure 6. Google Maps – Hokka Cafe: rating, opening hours and review-based tips; no live crowd information



Figure 7. Gym website (spor.yildiz.edu.tr): facility description and membership info; no real-time crowd information

3. Features

The app has six must-have features, each mapped to a course topic.

#	Feature	Course topic	What the user does
1	Moving between screens: venue list, venue detail, report sheet, sign in / sign up, profile	Navigation	Taps a venue in the list to open its detail page, opens the report sheet from there, switches to the profile from the bottom bar
2	Sign-in with a university email and live occupancy	Firebase (Auth + Firestore)	Signs up with a @std.yildiz.edu.tr address and verifies it; a report made on one phone appears on every other phone instantly
3	Last known status without internet	Local DB (Hive)	Sees each venue's last status with an "updated 12 min ago" label when offline
4	Weather warning	API / JSON (Open-Meteo)	Sees a note above the list that venues with outdoor seating are not recommended while it rains
5	Animated occupancy indicator	Animation	When a venue's status changes, the indicator's color and fill bar change smoothly
6	Each screen's logic lives in its own ViewModel	MVVM + Riverpod	(Invisible) Screens only draw; calculations and data access happen in ViewModels

Rules

- **Viewing is open:** No sign-in is needed to see how crowded venues are.
- **Reporting needs a verified university email:** Only verified @std.yildiz.edu.tr accounts can report. The rule is enforced both in the app and in Firestore security rules.
- **You must be at the venue:** The report button works only within 100–150 m of the venue (GPS). For the presentation, a demo mode in settings turns this check off.
- **Rate limit:** One person can report on the same venue at most once every 15 minutes.
- **Three levels:** Empty, Moderate, Full.
- **Opening hours:** Outside opening hours a venue shows "Closed", moves to the bottom of the list and cannot receive reports. The dining hall is open 11:30–14:30 and 17:00–19:15.
- **Interface language:** The app's interface is in Turkish because its users are YTU students. All documentation and code are in English.

Venues

Seven venues on Davutpaşa Campus: Hokka Cafe, Kayıntı Cafe, Büyük Ev Cafe, Yıldız Cafe, Yirmibirco, YTU Dining Hall and YTU Gym. The list is fixed; users cannot add venues. The dining hall's lower and upper floors are calculated separately, so the database holds 8 records; the list shows both floors on a single card with two rows.

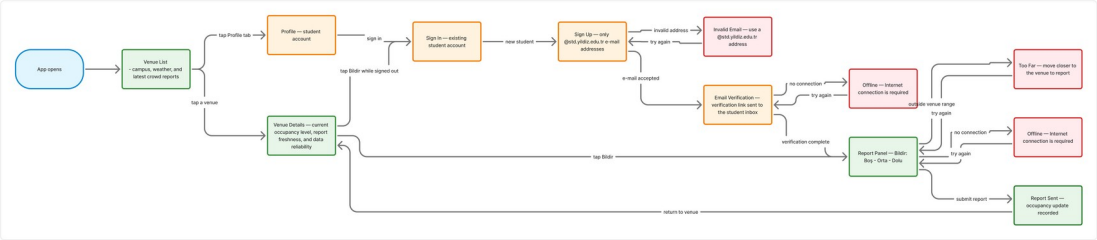
Later versions

An hourly busyness chart (“usually full at 12:30”) is not part of the MVP: it needs weeks of collected data to be meaningful.

4. Architecture

The app consists of 7 screens, 2 Firestore collections and MVVM layers; state is managed with Riverpod.

4.1 Screen map



The app opens directly on the venue list, because viewing needs no sign-in. The sign-in screen is reached only when the user wants to report, or from the profile. Mockup: Figma · 12 screens in docs/ (mockup_1_venue_list.png … mockup_12_report_sent.png). Screens 9–12 show the rules in action: offline mode, a closed venue, the distance check and the 15-minute limit after a report.

4.2 Data model

Venue (Firestore venues, 8 records)

Field	Type	Example	Used for
name	string	“Hokka Cafe”	Card title
type	string	cafe / dining / gym	Filters
group	string (optional)	“dining”	Showing both dining hall floors on one card
photoUrl	string	image URL	Card image
lat, lng	number	41.027, 28.889	GPS check
radiusM	number (m)	120	GPS check
halfLifeMin	number (min)	10	Occupancy calculation
hasOutdoor	boolean	true	Weather warning
openingHours	list of time ranges	[11:30–14:30, 17:00–19:15]	“Closed” status

Report (Firestore reports, one record per report)

Field	Type	Example
venueId	string	“hokka”
userId	string	Firebase Auth user id
level	number	0 (empty) / 50 (moderate) / 100 (full)
createdAt	timestamp	10.11.2026 12:31

User: No separate collection. Name, email and verification status are kept by Firebase Auth (name as the display name); the 15-minute rule is checked against the reports collection. **On the phone (Hive):** Each venue’s last calculated status and when it was calculated. This is shown when offline.

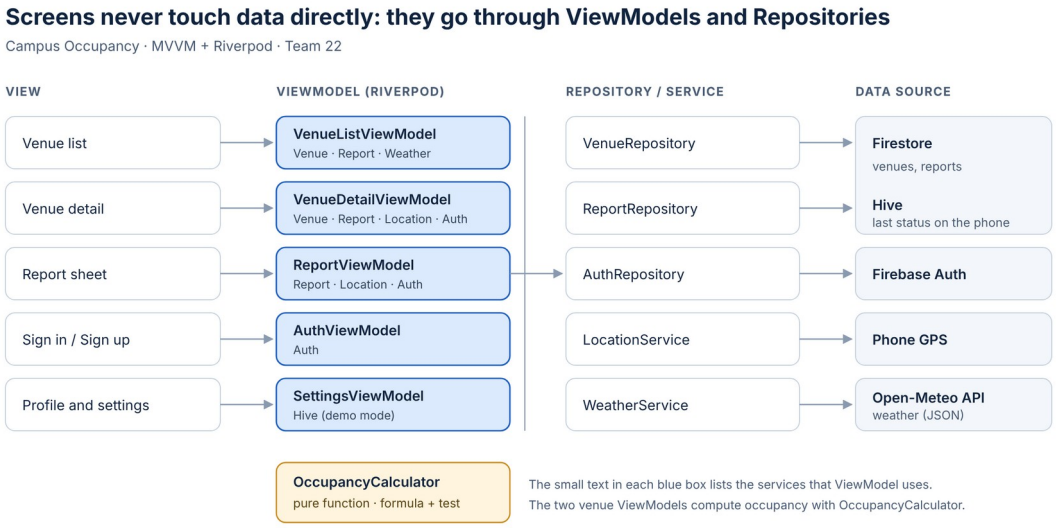
How occupancy is calculated

A venue’s status is not stored; it is calculated from recent reports every time. This way an old “full” never gets stuck on the screen.

1. Each report gets a weight based on its age: weight = 0.5 ^ (minutes since report / half-life). The half-life is 7 minutes for the dining hall, 10 for cafés and 20 for the gym. Reports older than about 4.5 half-lives are ignored (30 / 45 / 90 min).
2. The occupancy percentage is the weighted average: Σ(weight × level) / Σ(weight).
3. Label: 0–35% Empty, 35–65% Moderate, 65–100% Full. The percentage is shown only on the detail screen.
4. If the weights add up to less than 1.5 the card says “low confidence”; with no reports it says “no data”. Outside opening hours it says “Closed”.

Example: Three people reported “empty” 25 minutes ago and one reported “full” 2 minutes ago (café, half-life 10 min). The weights are 3 × 0.177 and 0.871; occupancy = 87.1 / 1.401 = **62%** → **Moderate**, with low confidence because the weights sum to 1.40. A plain average would give 25% (Empty) and miss the most recent “full” report.

4.3 MVVM diagram



Layer	Role	In this app
View	Only draws and passes user actions to the ViewModel	Venue list, Venue detail, Report sheet, Sign in / Sign up, Profile
ViewModel	Makes decisions and holds the screen's state	VenueListViewModel, VenueDetailViewModel, ReportViewModel, AuthViewModel, SettingsViewModel
Repository / Service	Knows where the data comes from	VenueRepository, ReportRepository, AuthRepository, LocationService, WeatherService
Data source	The data itself	Firestore, Firebase Auth, Hive, GPS, Open-Meteo

The occupancy formula lives in a pure function (OccupancyCalculator) outside the layers, so the example above can become an automated unit test.

The path of a report: The user taps "Full" → ReportViewModel checks email verification, distance, opening hours and the 15-minute rule → ReportRepository writes the record to Firestore → Firestore delivers the new report to every phone → VenueListViewModel on each phone recalculates occupancy → the cards update.

4.4 Why Riverpod

We chose Riverpod over Provider:

1. **It connects directly to live data:** StreamProvider binds the Firestore report stream to the screen in a single line. This live stream is the core of the app.
2. **ViewModels can use each other easily:** The venue detail ViewModel needs both the location service and the sign-in state. In Riverpod this needs no BuildContext.
3. **Errors are caught early:** Provider's "ProviderNotFound" error appears only at runtime; Riverpod catches similar mistakes at compile time.

5. Plan and Risks

Each member owns a layer: Oğuzhan builds the data and logic (repositories, ViewModels, Firebase), Sıla and Rabia build the screens (views) from the Figma mockup. At the end of each week a working version is merged into main. The final presentation date is not yet announced; the plan assumes six working weeks and a preparation week.

Weekly tasks

Week	Oğuzhan Arslan	Sıla	Rabia
1 (Nov 16–22)	Flutter project, MVVM folder structure, Firebase project and Auth	Flutter basics; app theme, colors and fonts	Flutter basics; VenueCard widget
2 (Nov 23–29)	Firestore venues / reports, VenueRepository, ReportRepository	Venue list screen	Venue detail screen
3 (Nov	OccupancyCalculator with unit tests,	Sign in, sign up and	Report sheet and

Week	Oğuzhan Arslan	Sıla	Rabia
30–Dec 6)	VenueListViewModel	email verification screens	status animations
4 (Dec 7–13)	Location check, 15-minute rule, opening hours, Firestore security rules	Profile and settings (demo mode)	Weather strip (Open-Meteo)
5 (Dec 14–20)	Hive offline cache, WeatherService	Offline and closed states	“Too far” and “report sent” states
6 (Dec 21–27)	Integration, bug fixes, APK build	Testing on real phones; collecting real reports on campus	2-minute demo video
7	Presentation rehearsal (all members)		

Risks

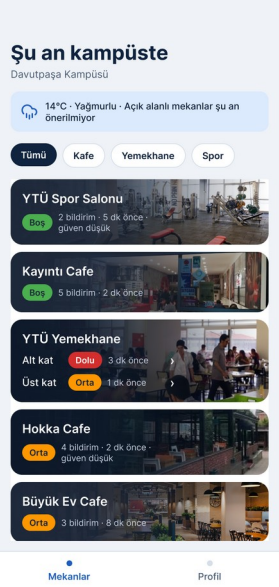
Risk	Impact	Mitigation
Few reports at the start (“cold start”)	Venues show “no data” and the app looks empty	One-tap reporting; “low confidence” label; team members report on campus during testing week
False reports	Wrong status shown	Verified university email, GPS check, 15-minute limit and weighted majority (see Section 4)
GPS is inaccurate indoors (20–50 m)	Users inside a venue cannot report	100–150 m radius per venue; demo mode for the presentation
University mail server delays or blocks verification emails	Users cannot finish sign-up	Test with team accounts in week 1; “resend email” button and spam-folder hint
Two members are new to Flutter	Screen work falls behind	Week 1 is for basics; screens are built from the existing Figma components; weekly pair-programming session
Firebase free-tier limits	Writes or reads are blocked	Only recent reports are queried (30–90 min windows); no images stored in Firebase
Exam weeks overlap with the plan	Weekly versions slip	Sunday 30-minute progress meeting; tasks are reassigned when someone falls behind

Use of AI

We used Claude for discussing the idea, planning the architecture, drafting parts of the report and the first version of the Figma mockup. Every member reviewed and can explain the part they are responsible for.

Appendix: Mockup screens

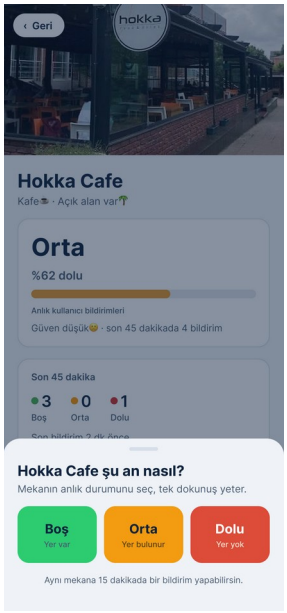
All 12 screens are in the Figma file and in the repository's docs/ folder. Screens 9–12 show the app's rules: offline mode, a closed venue, the distance check and the 15-minute limit after a report.



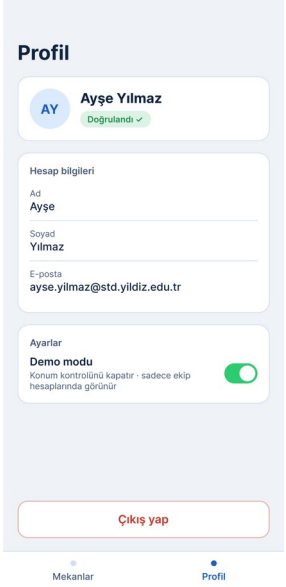
1 · Venue list



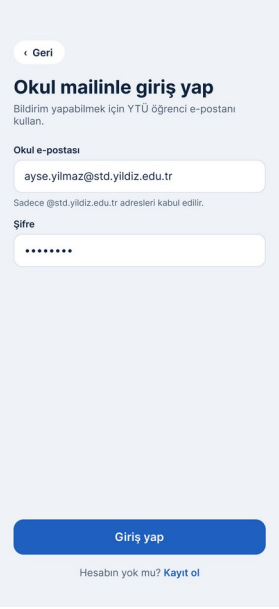
2 · Venue detail



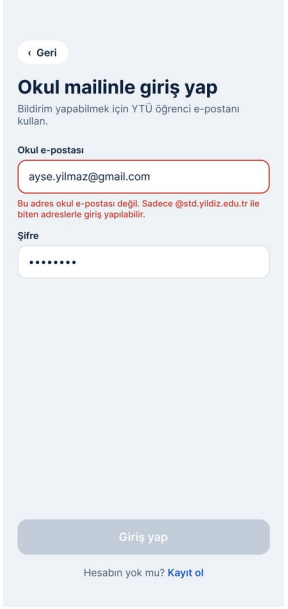
3 · Report sheet



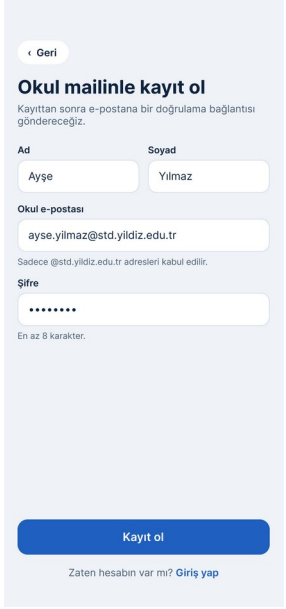
4 · Profile



5 · Sign in



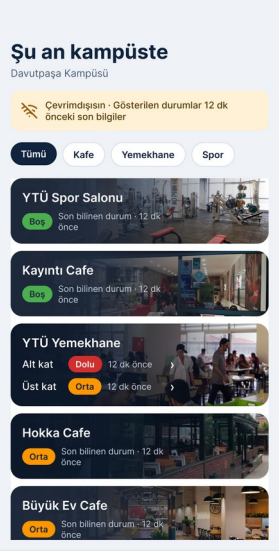
6 · Invalid email



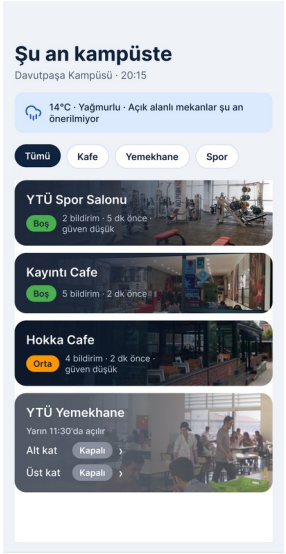
7 · Sign up



8 · Email verification



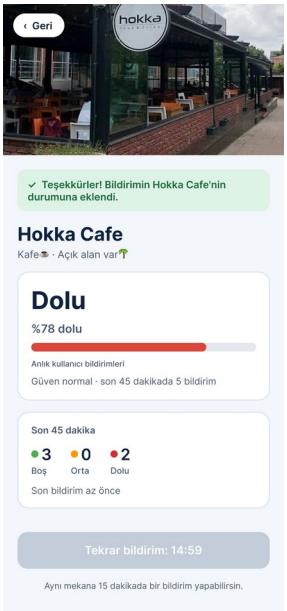
9 · Offline



10 · Closed venue



11 · Too far to report



12 · Report sent